

Jacobians in Other Frames

EGR 445/545

GVSU

plan for today

- inverse kinematics review example
- hints on LA 13
- debrief LA 11
- project time

Inverse Kin. Review Example

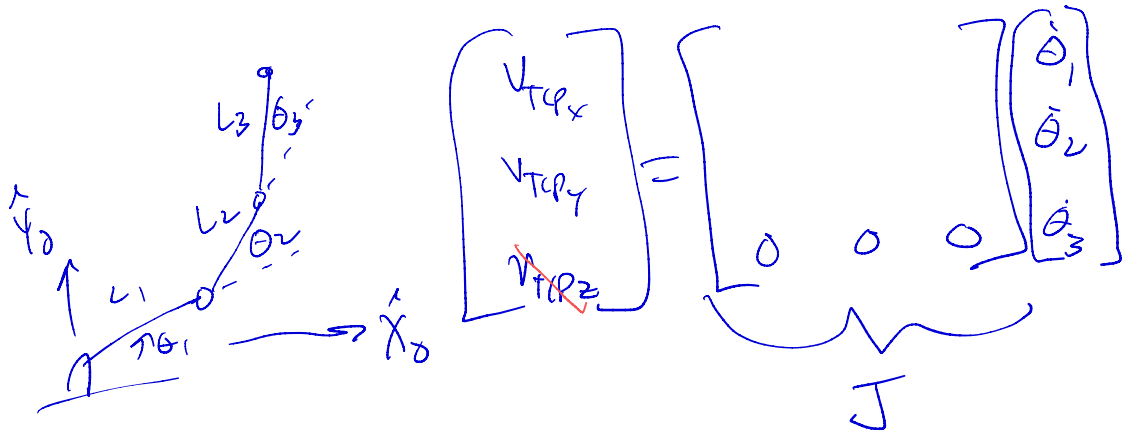
`review_inverse_kin_example.ipynb`

LA 13 Hint

Three link planar Jacobian:

if we use v_{tipz} as the third DOF, what will the bottom row of the Jacobian be?

v_{tipz} is ALWAYS 0
- it cannot be used as
an output velocity



$$\begin{bmatrix} v_{tipx} \\ v_{tipy} \\ \omega_{zlink3} \end{bmatrix} = \begin{bmatrix} J \end{bmatrix} \begin{bmatrix} \dot{\theta}_1 \\ \dot{\theta}_2 \\ \dot{\theta}_3 \end{bmatrix}$$

$\uparrow$
 $3\hat{\omega}_z$

Debrief LA 11

- why does this *almost* work?
 - what's the risk?

$${}^{w_{tip}}\mathbf{T}_{sym} = \begin{bmatrix} -\sin(\theta_4)\cos(\theta_5) & \sin(\theta_4)\sin(\theta_5) & \cos(\theta_4) & 9\cos(\theta_4) \\ \cos(\theta_4)\cos(\theta_5) & -\sin(\theta_5)\cos(\theta_4) & \sin(\theta_4) & 9\sin(\theta_4) \\ \sin(\theta_5) & \cos(\theta_5) & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

`c_th4 = T_wtip_num[0, 0]`

`s_th4 = T_wtip_num[1, 0]`

`th4 = np.arctan2(s_th4, c_th4) * rtd`

$$- \tan \theta_4 =$$

$$\frac{-\sin(\theta_4)\cos(\theta_5)}{\cos(\theta_4)\cos(\theta_5)}$$

? what if $\theta_5 = 90^\circ$?

? what if $\theta_5 < 0$?